

Direct Generation of a Somos–4 Sequence from an Algebraic Generating Function

Thomas Scheuerle

Abstract

Let

$$G(z) = \sum_{n \geq 0} g_n z^n$$

be a formal power series, and let

$$H_n = \det(g_{i+j})_{0 \leq i, j \leq n-1}, \quad H_0 = 1.$$

We consider the five-parameter family in which

$$H_1 = \lambda, \quad H_2 = m, \quad H_3 = r, \quad H_4 = \eta,$$

and the Hankel sequence satisfies the Somos–4 recurrence

$$H_{n+4}H_n = H_{n+3}H_{n+1} + \tau H_{n+2}^2.$$

We construct a quadratic algebraic generating function whose coefficient sequence has these initial Hankel determinants and whose Hankel transform lies in the family $A(1, \tau)$. The construction begins with a Stieltjes continued fraction, passes to a normalized form, and determines the coefficients by a paired-Hankel reconstruction. The resulting algebraic equation is then obtained by eliminating the continued-fraction tail. The two algebraic branches coalesce at the origin, and the desired branch is selected by the initial conditions. The essential point is that the construction removes this ambiguity by prescribing, in addition to the ordinary Hankel transform, a companion Hankel sequence obtained from the same Somos–4 orbit by a two-step shift.

Acknowledgments

I am grateful to Florian Lengyel for reading an earlier version of this manuscript and for pointing out important errors, and to Michael Somos for pointing out important suggestions for improvement.

1 Context and relation to earlier work

The present note is concerned with an inverse Hankel-realization problem. We start from prescribed initial determinants

$$(H_1, H_2, H_3, H_4) = (\lambda, m, r, \eta)$$

and a Somos–4 recurrence with parameter τ , and we construct a formal power series whose leading Hankel minors realize the corresponding $A(1, \tau)$ orbit. In addition, the construction fixes the once-shifted minors, and this paired-Hankel normalization determines the odd and even Stieltjes coefficients separately. The resulting alternating tail recurrence closes quadratically, and after restoring the initial levels one obtains an explicit algebraic equation for the generating function.

This is the inverse counterpart to the better-known forward problem: given a generating function, determine whether its Hankel transform satisfies a Somos recurrence. The latter has been studied, for example, by Xin, Barry, Wang–Zhang, and Liu–Wang–Zhang. The present construction takes the opposite direction and seeks a direct realization of prescribed Somos–4 Hankel data.

Seeking literature on the inverse problem, we find: Chang, Hu, and Xin construct Hankel solutions after a gauge normalization of adjacent terms [6]. Hone develops genus-one continued-fraction and Hankel formulae, including gauge factors and an additional moment parameter [14]. Chang and Liu treat generic bilateral initial conditions for quadratic orthogonal pairs and obtain algebraic series, coefficient recurrences, and Hankel representations [7]. These works provide the closest setting for the present construction.

The contribution of this note is a direct Stieltjes-fraction realization of generic $A(1, \tau)$ data in a particular positive-order paired-Hankel normalization. Here $H_0 = 1$ is the empty-determinant convention, while the recurrence starts at $n = 1$, so the four initial determinants remain independent. The ordinary and once-shifted determinant prescriptions recover the Stieltjes coefficients, and their alternating recurrence is compatible with a quadratic family of continued-fraction tails. Restoring the initial levels then gives the explicit quadratic equation recorded below.

The underlying tools come from the classical theory of continued fractions, moment sequences, and Hankel determinants; see Wall [24], Jones and Thron [16], Flajolet [8], Krattenthaler [17, 18], Layman [19], and French [10]. The quadratic transformation of Sulanke and Xin [22] and the related methods of Gessel and Xin [12] are particularly effective for structured Hankel identities. The wider Somos context includes the Laurent phenomenon and elliptic dynamics [11, 9, 23, 13, 15].

2 Hankel determinants and Somos parameters

Let

$$G(z) = \sum_{n \geq 0} g_n z^n$$

be a formal power series, and for $n \geq 1$ define its ordinary Hankel determinants by

$$H_n = \det(g_{i+j})_{0 \leq i, j \leq n-1}, \quad H_0 = 1. \quad (1)$$

The five parameters $\lambda, m, r, \eta, \tau$ have the following roles:

$$H_1 = \lambda, \quad H_2 = m, \quad H_3 = r, \quad H_4 = \eta, \quad (2)$$

and the Hankel sequence satisfies the Somos–4 recurrence

$$H_{n+4}H_n = H_{n+3}H_{n+1} + \tau H_{n+2}^2, \quad n \geq 1. \quad (3)$$

Thus τ is the second coefficient of the Somos family $A(1, \tau)$, whereas λ, m, r , and η select the initial Hankel determinants. The algebraic generating function producing these determinants is given in the next section. The construction makes one additional choice that is easy to overlook: together with $(H_n)_{n \geq 0}$, it fixes the companion Hankel sequence

$$\hat{H}_n = \det(g_{i+j+1})_{0 \leq i, j \leq n-1} = H_{n+2} \quad (n \geq 1). \quad (4)$$

Thus the companion is obtained by advancing the Somos–4 orbit by two indices without applying any additional normalization factor. Its first term is $\hat{H}_1 = g_1 = H_3 = r$, not $H_2 = m$. This second prescription is part of the definition of the selected coefficient sequence; it is not a consequence of the ordinary Hankel determinants alone. Section 9 explains why prescribing both sequences removes the degree of freedom discussed there.

3 Behind the construction: the Stieltjes fraction

The quadratic equation below may be obtained by first constructing a Stieltjes continued fraction whose Hankel determinants follow the prescribed Somos–4 orbit. Write

$$G(x) = \frac{c_0}{1 - \frac{c_1 x}{1 - \frac{c_2 x}{1 - \frac{c_3 x}{\ddots}}}}. \quad (5)$$

For this normalization, the ordinary Hankel determinants are [24, 16, 8]

$$H_n = c_0^n \prod_{j=1}^{n-1} (c_{2j-1} c_{2j})^{n-j}. \quad (6)$$

Thus

$$H_1 = c_0, \quad H_2 = c_0^2 c_1 c_2, \quad H_3 = c_0^3 c_1^2 c_2^2 c_3 c_4,$$

and similarly at higher orders.

It is useful first to remove the exponential scaling caused by H_1 . Set

$$a_1 = 1, \quad a_2 = \frac{m}{\lambda^2}, \quad a_3 = \frac{r}{\lambda^3}, \quad a_4 = \frac{\eta}{\lambda^4}. \quad (7)$$

Then $H_n = \lambda^n a_n$ for $1 \leq n \leq 4$, and the normalized sequence (a_n) obeys the same Somos–4 recurrence,

$$a_{n+4} a_n = a_{n+3} a_{n+1} + \tau a_{n+2}^2, \quad n \geq 1. \quad (8)$$

The powers of λ in (7) are essential: for nonunit $H_1 = \lambda$, the correct normalization is $a_j = H_j / \lambda^j$, not H_j / λ^{j-1} .

A convenient initial segment of the Stieltjes coefficients is

$$c_0 = \lambda, \quad c_1 = \lambda^2 a_3, \quad c_2 = \frac{a_2}{\lambda^2 a_3}, \quad (9)$$

$$c_3 = \frac{a_4}{a_2 a_3}, \quad c_4 = \frac{a_3^2}{a_2 a_4}, \quad c_5 = \tau \frac{a_2 a_3}{a_4} + \frac{a_2^2}{a_3}. \quad (10)$$

Equivalently, in the parameters used throughout this paper,

$$\begin{aligned} c_0 &= \lambda, & c_1 &= \frac{r}{\lambda}, & c_2 &= \frac{m}{\lambda r}, \\ c_3 &= \frac{\lambda \eta}{m r}, & c_4 &= \frac{r^2}{m \eta}, & c_5 &= \frac{\tau m r}{\lambda \eta} + \frac{m^2}{\lambda r}. \end{aligned} \quad (11)$$

The remaining coefficients are generated alternately by

$$c_{2k} = \frac{1}{c_{2k-2} c_{2k-1}}, \quad c_{2k+1} = c_{2k-2} + \frac{\tau}{c_{2k-1}} \quad (k \geq 3). \quad (12)$$

In particular,

$$c_6 = \frac{1}{c_4 c_5}, \quad c_7 = c_4 + \frac{\tau}{c_5}, \quad c_8 = \frac{1}{c_6 c_7}, \quad c_9 = c_6 + \frac{\tau}{c_7}.$$

Substitution in (6) gives $H_j = \lambda^j a_j$ for $1 \leq j \leq 4$. At the next order it gives

$$\frac{H_5}{\lambda^5} = \frac{a_3 a_4 c_5}{a_2} = a_2 a_4 + \tau a_3^2,$$

which is exactly the first Somos-4 step. The alternating recurrence (12) continues this mechanism.

The standard once-shifted Hankel product formula [18] for the Stieltjes fraction is

$$\hat{H}_n = (c_0 c_1)^n \prod_{j=1}^{n-1} (c_{2j} c_{2j+1})^{n-j} \quad (n \geq 1). \quad (13)$$

Together with (6), this formula gives a direct reconstruction result.

3.1 Paired-Hankel reconstruction

Let F be a field, let $H_0 = K_0 = 1$, and suppose $H_n K_n \neq 0$ for $n \geq 1$. There is a unique nondegenerate Stieltjes fraction of the form (5) whose ordinary and once-shifted Hankel determinants are H_n and K_n , respectively. Its coefficients are

$$c_0 = H_1, \quad c_{2n-1} = \frac{H_{n-1} K_n}{H_n K_{n-1}}, \quad c_{2n} = \frac{H_{n+1} K_{n-1}}{H_n K_n} \quad (n \geq 1). \quad (14)$$

Indeed, taking adjacent determinant ratios in (6) and (13) and telescoping alternately gives (14). Conversely, substitution of these coefficients into the two product formulae telescopes back to H_n and K_n . Nonvanishing makes every division legitimate and proves uniqueness in the S-fraction chart.

In the present construction $K_0 = 1$ and $K_n = H_{n+2}$ only for $n \geq 1$. Thus $c_1 = H_3/H_1$, $c_2 = H_2/(H_1 H_3)$, and, for $n \geq 2$,

$$c_{2n-1} = \frac{H_{n-1} H_{n+2}}{H_n H_{n+1}}, \quad c_{2n} = \frac{H_{n+1}^2}{H_n H_{n+2}}. \quad (15)$$

These identities recover (11) and (12). They also prove for every positive order that

$$\det(g_{i+j+1})_{0 \leq i, j \leq n-1} = H_{n+2} \quad (n \geq 1). \quad (16)$$

In particular, $g_1 = c_0 c_1 = r$. Thus the ordinary determinants fix only the products $c_{2j-1} c_{2j}$, whereas the companion determinants fix their splitting into odd and even coefficients. This gives the particular representative used throughout the paper. The formulas are classical in substance; their role here is to make the all-index determinant assertions and the qualified uniqueness statement explicit. The next step is to eliminate the recursively defined continued-fraction tail. The following isolated calculation explains why this elimination closes quadratically and hence provides the conceptual bridge to the explicit equation in Section 4.

3.2 Quadratic closure of the recurrent tail

For a sequence $(d_n)_{n \geq 1}$, write

$$T_n(x) = \frac{1}{1 - \frac{d_n x}{1 - \frac{d_{n+1} x}{1 - \ddots}}}, \quad T_n = \frac{1}{1 - d_n x T_{n+1}}. \quad (17)$$

The recurrence in (12) becomes transparent after the tail is started at c_4 . Put

$$d_n = c_{n+3}, \quad b = d_1 = c_4, \quad d = d_2 = c_5, \quad t = \tau. \quad (18)$$

Then $d_1 = b$, $d_2 = d$, $d_3 = 1/(bd)$, and thereafter

$$d_n = \frac{1}{d_{n-1} d_{n-2}} \quad (n \geq 3 \text{ odd}), \quad d_n = \frac{t}{d_{n-2}} + d_{n-3} \quad (n \geq 4 \text{ even}). \quad (19)$$

Thus the alternating recurrence can be studied independently of the first three levels of the original Stieltjes fraction.

Proposition 1 (Quadratic equation for the recurrent tail). *Let $T = T_1$ be the formal continued fraction defined by (17)–(19). Then $T(0) = 1$ and*

$$\mathcal{A}(x)T^2 + \mathcal{B}(x)T + \mathcal{C}(x) = 0, \quad (20)$$

where

$$\mathcal{A}(x) = (bt + 1)x - b((b + d)t + 1)x^2, \quad (21)$$

$$\mathcal{B}(x) = bdt x^2 + (-bt + b^2d + d^2b - 1)x - bd, \quad (22)$$

$$\mathcal{C}(x) = bd - bd^2x. \quad (23)$$

Proof. Removing one level of a Stieltjes fraction is a fractional-linear transformation,

$$T_{n+1} = \frac{T_n - 1}{d_n x T_n}. \quad (24)$$

Consequently, quadratic relations are preserved under tail extraction. More explicitly, if $Q_n(Y)$ is quadratic and $Q_n(T_n) = 0$, then a polynomial for the next tail is

$$Q_{n+1}(Y) = (1 - d_n x Y)^2 Q_n\left(\frac{1}{1 - d_n x Y}\right), \quad (25)$$

For the quadratic in (20), write $Q_{b,d}(x, Y) = \mathcal{A}(x)Y^2 + \mathcal{B}(x)Y + \mathcal{C}(x)$ and set

$$b' = \frac{1}{bd}, \quad d' = b + \frac{t}{d}.$$

A direct expansion gives the exact covariance identity

$$(1 - bx - dxZ)^2 Q_{b,d}\left(x, \frac{1 - dxZ}{1 - bx - dxZ}\right) = \frac{b^3 d^4 x^2}{bd + t} Q_{1/(bd), b+t/d}(x, Z). \quad (26)$$

For the recurrent tails,

$$T_1 = \frac{1 - dxT_3}{1 - bx - dxT_3}.$$

The recurrence (19) maps the parameter pair (b, d) for T_1 to $(1/(bd), b + t/d)$ for T_3 . Iterating (26) therefore shows, for every N , that $Q_{b,d}(x, T_1)$ is divisible by x^{2N} in $\mathbb{Q}(b, d, t)[[x]]$. Hence it vanishes identically. Finally, $Q_{b,d}(0, Y) = bd(1 - Y)$, so the formal branch satisfying $T_1(0) = 1$ is unique. $\square$

This proposition gives a short derivation of the algebraic nature of the full generating function. Let $T = T_4$ denote the tail of (5) beginning with c_4 . The first three levels are recovered by

$$T_3 = \frac{1}{1 - c_3 x T}, \quad T_2 = \frac{1}{1 - c_2 x T_3}, \quad T_1 = \frac{1}{1 - c_1 x T_2}, \quad G = c_0 T_1. \quad (27)$$

Each step is again a Möbius transformation. Substituting (27) into (20), followed by $c_0 = \lambda$ and the values in (11), therefore produces a quadratic polynomial in G . Clearing denominators and collecting powers of z yields exactly the polynomials in (31)–(38). This explains structurally why the recursively defined infinite fraction gives a quadratic algebraic generating function: the recurrence is the compatibility condition that keeps the continued-fraction tails inside a quadratic family, while the omitted initial levels only conjugate that family by fractional-linear maps.

4 The algebraic generating function

Define the polynomials $A(z), B(z), C(z)$ by

$$A(z) = \lambda\eta mr^2(\tau - \eta)z^3 \quad (28)$$

$$+ r\left(-(\lambda^2\eta m + m^2(m + r^2))\tau + (2m - 1)\lambda^2\eta^2 + \eta m^2(m + r^2) - \lambda r(m + r^2)\right)z^2 \quad (29)$$

$$+ \lambda\left(m^2r^2\tau - (m - 1)\lambda^2\eta^2 - m(m^2 + 2mr^2 - m - r^2)\eta + \lambda r^3\right)z \quad (30)$$

$$+ \lambda^2\eta m(m - 1)r, \quad (31)$$

$$B(z) = -\lambda^2\eta mr^2\tau z^3 \quad (32)$$

$$+ r\left((2\lambda^3\eta m + \lambda m^2(2m + r^2))\tau - (2m - 1)\lambda^3\eta^2 - \lambda\eta m^3 + \lambda^2r(2m + r^2)\right)z^2 \quad (33)$$

$$+ \lambda^2\left(-2m^2r^2\tau + 2(m - 1)\lambda^2\eta^2 + m(2m^2 + 2mr^2 - 2m - r^2)\eta - 2\lambda r^3\right)z \quad (34)$$

$$- 2\lambda^3\eta m(m - 1)r, \quad (35)$$

$$C(z) = -\lambda^2mr\left(\tau(\lambda^2\eta + m^2) + \lambda r\right)z^2 \quad (36)$$

$$+ \left(\lambda^3m^2r^2\tau + \lambda^4r^3 - \lambda^3\eta(m - 1)(\lambda^2\eta + m^2)\right)z \quad (37)$$

$$+ \lambda^4\eta m(m - 1)r. \quad (38)$$

The generating function is the formal-power-series branch of

$$A(z)G(z)^2 + B(z)G(z) + C(z) = 0 \quad (39)$$

4.1 Discriminant and residual genus-one curve

Define

$$J = \frac{\lambda\eta}{mr} + \frac{m^2}{\lambda r} + \frac{r^2}{m\eta} + \frac{\tau mr}{\lambda\eta}. \quad (40)$$

For the polynomials above, direct expansion and factorization give

$$B(z)^2 - 4A(z)C(z) = \lambda^4\eta^2m^2r^4z^2((1 - Jz + \tau z^2)^2 - 4z^3). \quad (41)$$

After removing the square factor and setting $X = z^{-1}$, the residual quadratic extension is birational to

$$E_J : \quad y^2 = (X^2 - JX + \tau)^2 - 4X. \quad (42)$$

Its smooth completion is a genus-one curve whenever

$$J^4\tau + J^3 - 8J^2\tau^2 - 36J\tau + 16\tau^3 - 27 \neq 0. \quad (43)$$

Indeed, the discriminant in X of the quartic on the right of (42) is 256 times the left-hand side of (43). The parameter J is also the first integral of the standard QRT dynamics for $\beta_n = H_{n-1}H_{n+1}/H_n^2$. The Somos-4 recurrence gives $\beta_{n-1}\beta_n^2\beta_{n+1} = \beta_n + \tau$; hence the map $(u, v) \mapsto (v, (v + \tau)/(uv^2))$ preserves $uv + u^{-1} + v^{-1} + \tau(uv)^{-1}$. Substitution of $u = \lambda r/m^2$ and $v = m\eta/r^2$ yields (40).

5 The two algebraic branches

With

$$D(z) = B(z)^2 - 4A(z)C(z),$$

the two algebraic branches may be represented by

$$G(z) = \frac{-B(z) \pm \sqrt{D(z)}}{2A(z)}. \quad (44)$$

When the constant term of A vanishes, the rationalized expressions

$$\frac{-2C(z)}{B(z) \mp \sqrt{D(z)}}$$

can be used instead. The sign in (44) cannot be fixed uniformly over the full parameter space.

Write

$$P(z, X) = A(z)X^2 + B(z)X + C(z).$$

The constant coefficients of (31)–(38) are

$$A_0 = \lambda^2 \eta m(m-1)r, \quad B_0 = -2\lambda^3 \eta m(m-1)r, \quad C_0 = \lambda^4 \eta m(m-1)r.$$

Consequently,

$$P(0, X) = A_0(X - \lambda)^2. \quad (45)$$

Both algebraic branches therefore coalesce at $X = \lambda$ when $z = 0$, and

$$\partial_X P(0, \lambda) = 2A_0\lambda + B_0 = 0. \quad (46)$$

The simple-root form of the formal implicit-function theorem is not applicable at the origin. When $m = 1$, the polynomial has a factor z , and the cancelled equation satisfies

$$\left. \frac{P(z, X)}{z} \right|_{z=0} = \lambda r^2(X - \lambda)((\lambda r + \tau - \eta)X - \lambda(\lambda r + \tau)).$$

Thus, generically, the second branch then has constant term $\lambda(\lambda r + \tau)/(\lambda r + \tau - \eta)$ rather than λ . In either case the desired branch is selected by the paired conditions $g_0 = \lambda$ and $g_1 = r$.

$$G(z) = \lambda + zF(z), \quad F(z) = \sum_{q \geq 0} f_q z^q, \quad f_q = g_{q+1}. \quad (47)$$

For

$$D_j = 2\lambda A_j + B_j, \quad E_j = \lambda^2 A_j + \lambda B_j + C_j,$$

where $C_3 = 0$, substitution into $P(z, G(z)) = 0$ gives

$$P(z, \lambda + zF) = \sum_{j=0}^3 E_j z^j + z \left(\sum_{j=0}^3 D_j z^j \right) F + z^2 \left(\sum_{j=0}^3 A_j z^j \right) F^2.$$

Equation (45) gives $E_0 = D_0 = 0$. An ordinary power-series branch additionally requires

$$E_1 = 0. \quad (48)$$

The next coefficient equation is

$$A_0 f_0^2 + D_1 f_0 + E_2 = 0. \quad (49)$$

The quadratic in (49) factors as

$$A_0 f_0^2 + D_1 f_0 + E_2 = \lambda^2 \eta m r (f_0 - r)((m-1)f_0 - m r). \quad (50)$$

Thus the prescribed value $f_0 = g_1 = r$ selects the required branch. When $m \neq 1$, the other slope is $m r / (m - 1)$.

6 The two-step companion

Let $G_{0;\ell,m,r,n,t}(x) = \sum_{q \geq 0} g_q x^q$ denote the selected solution. Its Hankel determinants satisfy

$$(H_1, H_2, H_3, H_4) = (\ell, m, r, n), \quad H_{q+4}H_q = H_{q+3}H_{q+1} + tH_{q+2}^2. \quad (51)$$

The next two values are

$$H_5 = \frac{mn + tr^2}{\ell}, \quad (52)$$

$$H_6 = \frac{rH_5 + tn^2}{m} = \frac{r(mn + tr^2) + t\ell n^2}{\ell m}. \quad (53)$$

Remove the first coefficient and shift the remaining series to the origin:

$$\hat{G}(x) = \frac{G_{0;\ell,m,r,n,t}(x) - g_0}{x} = \sum_{q \geq 0} g_{q+1} x^q. \quad (54)$$

Its Hankel transform is

$$\det(g_{i+j+1})_{0 \leq i,j \leq q-1} = H_{q+2} \quad (q \geq 1). \quad (55)$$

It therefore begins with

$$r, \quad n, \quad \frac{mn + tr^2}{\ell}, \quad \frac{r(mn + tr^2) + t\ell n^2}{\ell m}, \dots$$

This is exactly the two-step advanced orbit $(H_{q+2})_{q \geq 1}$. It therefore satisfies the same recurrence with parameter t , starting at positive order $q \geq 1$. Equivalently, it has the initial parameter tuple

$$\left(r, n, \frac{mn + tr^2}{\ell}, \frac{r(mn + tr^2) + t\ell n^2}{\ell m}, t \right). \quad (56)$$

The assertion is an identity of Hankel transforms, not necessarily of the underlying coefficient sequences.

7 A recurrence for the coefficients

Using the notation of (47), division of the expanded quadratic equation by z^2 yields

$$A(z)F(z)^2 + (D_1 + D_2z + D_3z^2)F(z) + E_2 + E_3z = 0. \quad (57)$$

The branch condition is

$$A_0r^2 + D_1r + E_2 = 0, \quad (58)$$

and the branch is simple precisely when

$$2A_0r + D_1 \neq 0. \quad (59)$$

Equivalently,

$$(2A_0r + D_1)^2 = D_1^2 - 4A_0E_2. \quad (60)$$

Define

$$U_q = \sum_{k=0}^q f_k f_{q-k}, \quad U_q = 0 \quad (q < 0). \quad (61)$$

Coefficient extraction from (57) gives

$$\sum_{j=0}^3 A_j U_{q-j} + D_1 f_q + D_2 f_{q-1} + D_3 f_{q-2} + E_{q+2} = 0, \quad (62)$$

where $f_q = 0$ for $q < 0$ and $E_j = 0$ for $j > 3$. For $q \geq 1$,

$$U_q = 2r f_q + \sum_{k=1}^{q-1} f_k f_{q-k}. \quad (63)$$

Under condition (59), the coefficients are therefore generated recursively by

$$f_q = -\frac{A_0 \sum_{k=1}^{q-1} f_k f_{q-k} + \sum_{j=1}^3 A_j U_{q-j} + D_2 f_{q-1} + D_3 f_{q-2} + E_{q+2}}{2A_0 r + D_1}. \quad (64)$$

This quadratic convolution recurrence follows directly from the algebraic generating-function equation. On the selected branch,

$$2A_0 r + D_1 = -\lambda^2 \eta m r^2, \quad (65)$$

so condition (59) holds throughout the nondegenerate S-fraction chart. Because F is algebraic of degree at most two, it is D-finite. Consequently, the same coefficient sequence also satisfies a fixed-width linear recurrence with polynomial coefficients in the index.

8 Base field and nondegeneracy

The generic construction is over the rational-function field $\mathbb{Q}(\lambda, m, r, \eta, \tau)$. For scalar specializations, the initial conditions require at least

$$\lambda m r \eta (m \eta + \tau r^2) \neq 0,$$

together with the nonvanishing of every later Hankel quantity that occurs as a denominator. Accordingly, every reconstruction and uniqueness statement in this paper is understood within this nondegenerate S-fraction chart.

9 Nonuniqueness of the coefficient sequence

The ordinary Hankel transform does not, in general, determine a coefficient sequence. If only (H_n) is prescribed, the factorization $c_{2j-1} c_{2j}$ visible in (6) leaves room for variation in the individual Stieltjes coefficients. For this reason a five-parameter algebraic generating function should not be read as the unique moment representation attached to an ordinary Somos-4 Hankel orbit.

The construction in this paper removes this ambiguity by imposing the additional companion condition (4). In other words, we prescribe two product sequences: the first is the Somos-4 sequence (H_n) , and the second is the same sequence advanced by two determinant indices. Together with $c_0 = \lambda$, these two prescriptions determine the odd and even Stieltjes coefficients separately through (10) and (12). Hence the algebraic generating function in Section 4 is the fully specified representative selected by those two Hankel-level requirements. This choice is already built into the initial Stieltjes coefficients above; making it explicit here clarifies how the apparent degree of freedom is eliminated.

Other moment sequences can still share the first ordinary Hankel transform when the companion condition is not imposed. For example, the binomial transform, a standard Hankel-transform-preserving operation [19, 10],

$$g'_q = \sum_{k=0}^q \binom{q}{k} g_k, \quad (66)$$

has generating function

$$G'(z) = \frac{1}{1-z} G\left(\frac{z}{1-z}\right)$$

and preserves the Hankel determinants. Scalar and exponential rescalings of coefficient sequences produce explicit normalization factors in the determinants, while affine changes of the underlying variable give further related moment functionals. Equality of Hankel transforms must therefore not be replaced by equality of generating functions without an independent argument.

Disclosure on authoring and verification tools

The text of this manuscript was revised and partially rewritten with the assistance of AI-based language tools. The purpose was to improve clarity and place the mathematical material into proper $\text{\LaTeX}$ while preserving the technical content and the mathematical claims. The author remains responsible for the final text, all statements, and the correctness of the formulas.

AI-assisted workflows were also used to draft and test PARI/GP scripts for verifying the identities and formulas appearing in this paper. PARI/GP merits explicit recognition as a powerful and highly effective tool for exact symbolic and arithmetic verification, and it played a useful role in checking the computations underlying the present work.

References

- [1] P. Barry, Generalized Catalan numbers, Hankel transforms and Somos–4 sequences, *Journal of Integer Sequences* **13** (2010), Article 10.7.2.
- [2] P. Barry, On a generalization of the Narayana triangle, *Journal of Integer Sequences* **14** (2011), Article 11.4.5.
- [3] P. Barry, Riordan–Bernstein polynomials, Hankel transforms and Somos sequences, *Journal of Integer Sequences* **15** (2012), Article 12.8.2.
- [4] P. Barry, Riordan pseudo-involutions, continued fractions and Somos–4 sequences, *Journal of Integer Sequences* **22** (2019), Article 19.6.1; arXiv:1807.05794.
- [5] P. Barry, Conjectures on Somos–4, 6 and 8 sequences using Riordan arrays and the Catalan numbers, arXiv:2211.12637 [math.CO], 2022.
- [6] X.-K. Chang, X.-B. Hu, and G. Xin, Hankel determinant solutions to several discrete integrable systems and the Laurent property, *SIAM Journal on Discrete Mathematics* **29** (2015), no. 1, 667–682.
- [7] X.-K. Chang and J. Liu, Hankel determinants from quadratic orthogonal pairs for hyperelliptic functions and their applications, arXiv:2603.11670v4 [nlin.SI], 2026; to appear in the *Journal of the London Mathematical Society*.
- [8] P. Flajolet, Combinatorial aspects of continued fractions, *Discrete Mathematics* **32** (1980), 125–161.

- [9] S. Fomin and A. Zelevinsky, The Laurent phenomenon, *Advances in Applied Mathematics* **28** (2002), 119–144.
- [10] C. French, Transformations preserving the Hankel transform, *Journal of Integer Sequences* **10** (2007), Article 07.7.3.
- [11] D. Gale, The strange and surprising saga of the Somos sequences, *The Mathematical Intelligencer* **13** (1991), no. 1, 40–42.
- [12] I. M. Gessel and G. Xin, The generating function of ternary trees and continued fractions, *Electronic Journal of Combinatorics* **13** (2006), Research Paper 53.
- [13] A. N. W. Hone, Elliptic curves and quadratic recurrence sequences, *Bulletin of the London Mathematical Society* **37** (2005), 161–171.
- [14] A. N. W. Hone, Continued fractions and Hankel determinants from hyperelliptic curves, *Communications on Pure and Applied Mathematics* **74** (2021), no. 11, 2310–2347.
- [15] A. N. W. Hone and C. Swart, Integrality and the Laurent phenomenon for Somos–4 sequences, *Mathematical Proceedings of the Cambridge Philosophical Society* **145** (2008), 65–85.
- [16] W. B. Jones and W. J. Thron, *Continued Fractions: Analytic Theory and Applications*, Encyclopedia of Mathematics and its Applications, vol. 11, Addison–Wesley, 1980.
- [17] C. Krattenthaler, Advanced determinant calculus, *Séminaire Lotharingien de Combinatoire* **42** (1999), Article B42q.
- [18] C. Krattenthaler, Advanced determinant calculus: a complement, *Linear Algebra and its Applications* **411** (2005), 68–166.
- [19] J. W. Layman, The Hankel transform and some of its properties, *Journal of Integer Sequences* **4** (2001), Article 01.1.5.
- [20] F. Liu, Y. Wang, and Z. Zhang, Hankel transform and (α, β) Somos–4 sequences, arXiv:2608.08703 [math.CO], 2026, <https://arxiv.org/abs/2608.08703>.
- [21] V. Ovsienko and E. Pedon, Continued fractions for q -deformed real numbers, $\{-1, 0, 1\}$ -Hankel determinants, and Somos–Gale–Robinson sequences, *Advances in Applied Mathematics* **162** (2024), Article 102788.
- [22] R. A. Sulanke and G. Xin, Hankel determinants for some common lattice paths, *Advances in Applied Mathematics* **40** (2008), 149–167.
- [23] A. J. van der Poorten and C. S. Swart, Recurrence relations for elliptic sequences: every Somos–4 is a Somos– k , *Bulletin of the London Mathematical Society* **38** (2006), 546–554.
- [24] H. S. Wall, *Analytic Theory of Continued Fractions*, Van Nostrand, New York, 1948.
- [25] Y. Wang and Z. Zhang, A sufficient condition for (α, β) Somos–4 Hankel determinants, *Discrete Mathematics* (2024); arXiv:2305.05995.
- [26] G. Xin, Proof of the Somos–4 Hankel determinants conjecture, *Advances in Applied Mathematics* **42** (2009), 152–156.